

Reimplementing DreamBooth – Final Report

1. Introduction

Text-to-image diffusion models can generate high-quality images from natural language prompts, but they typically cannot generate a specific instance of a subject. For example, a model can generate an image of a professor, but will likely struggle to generate *Kilian Weinberger*. DreamBooth addresses this problem by taking a small set of images of a specific subject (~3-5 images) and fine-tunes a pretrained diffusion model so that a rare identifier token like “[V]” becomes associated with that subject. The fine-tuned model can then generate the subject in novel contexts, poses, styles, lighting, etc. It also introduces a new class-specific prior preservation loss to reduce language drift, where the model over-associates the class noun with the training subject, losing diversity. We reimplemented DreamBooth and extended beyond the original paper by integrating with LoRA and experimenting with one-shot tasks across multiple models.

2. Chosen Result

We focused on the paper’s main result: fine-tuning a pretrained text-to-image model on a few subject images that can generate the same subject in *new* contexts without overfitting the general class to the specific instance (i.e., fine-tuning on my dog while preserving the ability to generate any dog). In the original paper, this includes implementing the fine-tuning pipeline in Figure 3, the recontextualization examples in Figure 7, and the prior preservation loss ablation in Table 3, Figure 6, and Equation 2. These results were chosen because they highlight the key balance in DreamBooth – the model must learn the specific subject well enough for high-quality image generation, but not so strongly that it collapses diversity. On top of this, we also conducted the same LoRA-based experimentation mentioned above to determine whether a different fine-tuning regimen could make a meaningful difference in output quality.

3. Methodology

Our implementation followed the DreamBooth fine-tuning setup. For each subject, we used a small set of images of a specific subject and paired them with prompts in the general form of “a [V] [class noun]”. The rare token denotes the specific subject, while the class noun allows the model to use its existing concepts of the general class, i.e., dogs, cats, backpacks, vases, etc. We used the official DreamBooth data and additional personal images to test whether the model generalized beyond the original benchmarks.

We experimented with four pretrained models: Stable Diffusion XL, 1.5, 3M, and Flux2 4B. Additionally, we implemented two adaptation strategies: one was a full DreamBooth-style fine-tuning method that updated the model’s original weights. The second used LoRA, which trained low-rank matrices instead of the original model weights. As LoRA greatly reduced the memory load (28GB to 15GB), we could use larger learning rates and batch sizes. All training and benchmarks were done on an Nvidia RTX 5090.

For both full fine-tuning and LoRA, we compared runs with and without prior preservation loss (PPL). When using PPL, we generated 1000 class images from the base model using prompts such as “a dog”. When training, the specific instance reconstruction loss teaches the model the target subject, while the PPL encourages the model to continue generating diverse images of the broader class. By testing with and without the PPL, we can test the paper’s claim that the PPL mitigates language drift and preserves diversity. We also tested the model’s one-shot ability (one subject image) as a novel and harder setting.

We evaluated generations using metrics and visual inspection. The metrics were:

- **PRES↓**: prior preservation metric; smaller means less collapse of the prior to the learned subject.
- **DIV↑**: image diversity; higher means more variation across generated samples.
- **DINO & CLIP-I↑**: subject fidelity metrics; compares generated images to the reference subject.
- **CLIP-T↑**: text fidelity metric; compares generated images to the prompt.

4. Results and Analysis

We successfully replicated DreamBooth's behavior with both vanilla and LoRA fine-tuning. Qualitatively, the models with PPL were able to generate learned subjects in new contexts and poses, even in a one-shot scenario (albeit less reliably with less novel poses, etc.). Additionally, we observed that LoRA produced the highest-quality images of the special subject and created the most novel poses for our test subjects.

Method	PRES↓	DIV↑	DINO↑	CLIP-I↑	CLIP-T↑
DreamBooth original	0.664	0.371	0.712	0.828	0.306
DreamBooth original w/ PPL	0.493	0.391	0.684	0.815	0.308
Full model, ours	0.848	0.631	0.875	0.909	0.302
Full model, ours w/ PPL	0.419	0.654	0.853	0.910	0.304
LoRA model, ours	0.847	0.671	0.865	0.920	0.291
LoRA model, ours w/ PPL	0.372	0.702	0.857	0.912	0.294

The strongest trend was the inclusion of PPL, which matches the original paper's conclusion that PPL counteracts language drift and improves diversity. In fact, our models consistently outperform those in the original paper, except for CLIP-T. However, metrics are hard to compare as the original authors used different models, epochs, encoders, etc. We also observed that PPL slightly reduces the DINO score, suggesting a tradeoff between preserving the broader class prior and reducing subject memorization.

LoRA achieved our best PRES, DIV, and CLIP-I scores while producing similarly competitive results on DINO and CLIP-T, demonstrating that parameter-efficient tuning can effectively reproduce DreamBooth. Additionally, in combination with our qualitative assessment of LoRA-generated images, we theorize that LoRA can even slightly improve training by slightly reducing the model's tendency to overfit to the subject, due to its inherent low-rank structure, forcing it to learn the subject rather than memorize it.

We also observed three primary failure modes. First, some generations preserved the correct shape but changed key details, such as color, suggesting that the model only learned part of the subject identity. Second, sometimes the model simply did not learn the subject; this may be because it did not bind the subject to the rare identifier token. Third, strong pretrained priors sometimes dominated the learned subject, as in artistic prompts, where the style took precedence over the specific object.

5. Reflection

We learned that while DreamBooth is simple in concept, it is somewhat delicate in practice. The rare-token prompt and fine-tuning objectives are straightforward, but the quality of the generated output highly depends on the model, the number of subject images, the prompt design, the learning rate, the number of training steps, etc. If the model is undertrained, it may ignore the rare token and only generate a generic instance. If overtrained, it may memorize the input images, damaging the class prior.

Another important takeaway was the importance of prior preservation. Without PPL, the model learned the subject more aggressively, but at a great cost in diversity and class-level generalization. With PPL, the model better preserved the class prior, and generated images were more diverse; however, at the cost of rare failures that prevented the model from learning the subject identity.

Given more time, we would like to further investigate the failure modes, especially when the PPL prevents the model from learning aspects of the specific subject, and removing it would otherwise make it work. Additionally, we would like to further experiment with improving one-shot fine-tuning's diversity and fidelity. Finally, we would like to see if the model can create *novel combinations* of unique subjects.

6. References

- Ruiz, N., Li, Y., Jampani, V., Pritch, Y., Rubinstein, M., & Aberman, K. (2023). *DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation*. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 22500–22510.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2022). *LoRA: Low-Rank Adaptation of Large Language Models*. International Conference on Learning Representations (ICLR).